

Linear Maps

Chapter 1 studied vector spaces one at a time. This chapter studies the functions between them that respect the structure: **linear maps**, the functions that send sums to sums and scalar multiples to scalar multiples. Differentiation of polynomials, the trace of a matrix, a rotation of the plane and multiplication by a matrix are all linear maps, and one theory covers them all.

Two subspaces measure how far a linear map is from being injective and surjective: its kernel and its image. The Rank–Nullity Theorem ties their dimensions together, and turns many questions about maps into counting. Then we choose bases and write every linear map as a matrix. This explains why matrix multiplication is defined the way it is, and it raises the central question of the rest of the book: how does the matrix change when the bases change, and how simple can we make it?

What you need. From Chapter 1, bases and dimension (Definition 1.5.1, Definition 1.6.1), basis extension (Theorem 1.6.5) and counting instead of checking (Theorem 1.6.4). From Chapter 2, the null space and column space of a matrix (Definition 2.5.3, Definition 2.5.1), its rank (Definition 2.5.4, Theorem 2.5.7) and the invertible matrix theorem (Theorem 2.4.4). From Chapter 0, quotient sets and well-defined functions on them (Theorem 0.5.3), for the section on quotient spaces.

3.-1.1 Roadmap

- **Linear maps.** The definition, a gallery of examples and non-examples, and the fact that a linear map is determined by its values on a basis.
- **Kernel and image.** The two subspaces attached to every linear map, and the kernel test for injectivity.
- **The Rank–Nullity Theorem.** $\dim V = \text{nullity } T + \text{rank } T$, and what counting can decide.
- **The algebra of linear maps.** Adding, scaling and composing maps; polynomials in an operator.
- **Isomorphisms.** When two spaces are “the same”, and why dimension alone decides it.
- **The matrix of a linear map.** Coordinates turn maps into matrices and composition into matrix multiplication.
- **Change of basis and similarity.** How the matrix of a map depends on the chosen bases.
- **Rank, equivalence and factorizations.** With free choice of bases on both sides, every matrix becomes $\begin{pmatrix} \mathbf{I}_r & 0 \\ 0 & 0 \end{pmatrix}$.
- **Products and quotient spaces.** Building new spaces from old ones, and the isomorphism theorems.
- **Projections and the trace.** Direct sums seen through operators, and a basis-free trace.

3.-1.2 Named moves

- **Injective via the kernel.** To show T is injective, show $T\mathbf{v} = \mathbf{0}$ forces $\mathbf{v} = \mathbf{0}$.
- **Count with Rank–Nullity.** When $\dim V = \dim W$, injective, surjective and bijective are the same; check the easiest one.
- **Basis of the kernel, then extend.** The proof pattern behind Rank–Nullity, the same one that proved the dimension formula for sums.
- **Abstract space $\rightarrow$ write down a matrix.** Choose a basis that makes the map simple, then compute with its matrix.

- **The change-of-basis square.** Go around the square instead of across it: $[T]_{\mathcal{B}'}^{\mathcal{B}'} = \mathbf{P}^{-1}[T]_{\mathcal{B}}^{\mathcal{B}}\mathbf{P}$.
- **Check well-definedness on cosets.** Every operation on a quotient space, and every map out of one, starts with this check.